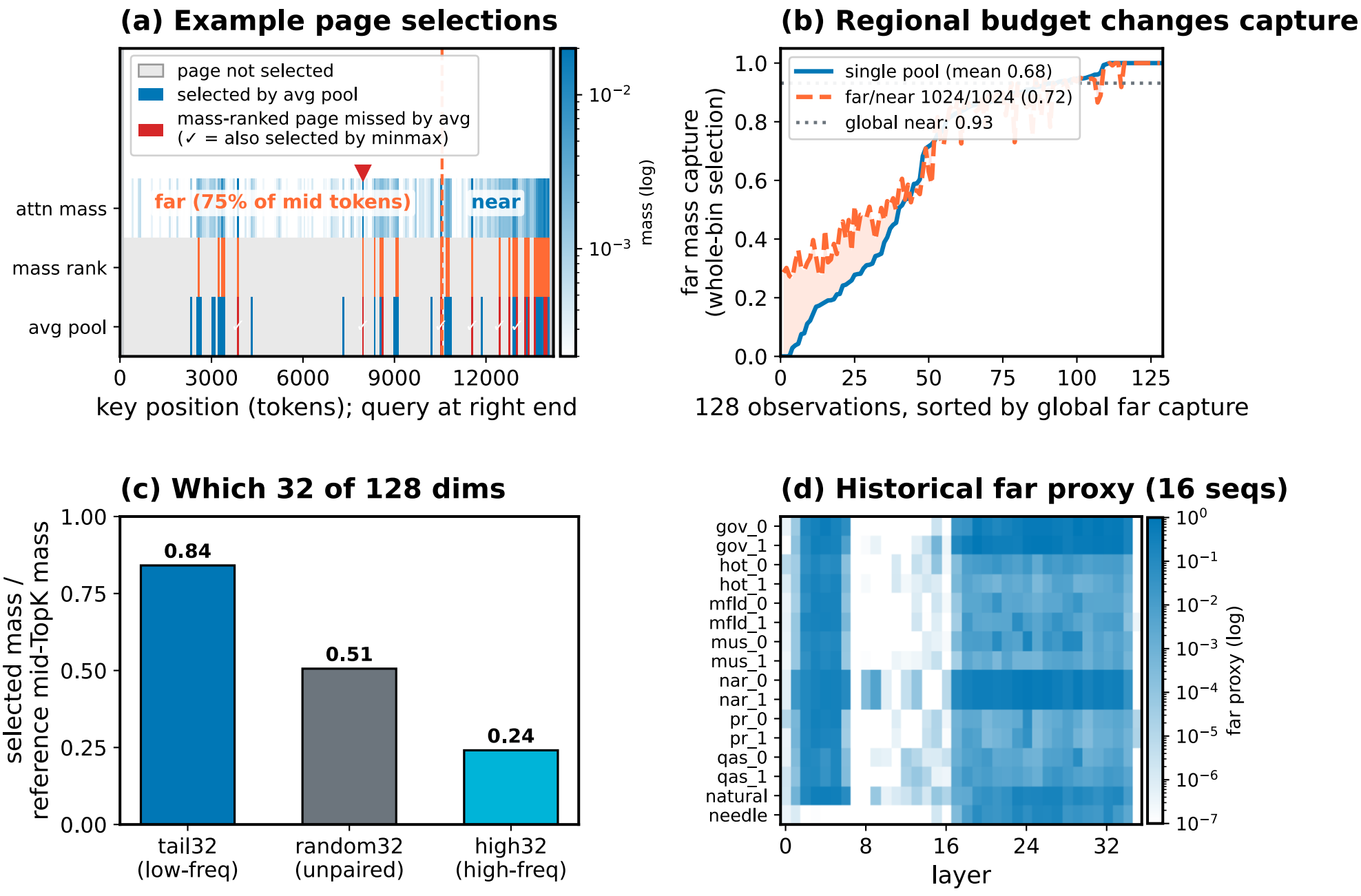

(a) Example page selections
page not selected
selected by avg pool
mass-ranked page missed by avg
(✓ = also selected by minmax)
attn mass
mass rank
avg pool
far (75% of mid tokens)
near
10^{-2}
10^{-3}
mass (log)
0
3000
6000
9000
12000
key position (tokens); query at right end
(b) Regional budget changes capture
single pool (mean 0.68)
far/near 1024/1024 (0.72)
global near: 0.93
far mass capture
(whole-bin selection)
0.0
0.2
0.4
0.6
0.8
1.0
0
25
50
75
100
125
128 observations, sorted by global far capture
(c) Which 32 of 128 dims
selected mass /
reference mid-TopK mass
0.00
0.25
0.50
0.75
1.00
0.84
0.51
0.24
tail32
(low-freq)
random32
(unpaired)
high32
(high-freq)
(d) Historical far proxy (16 seqs)
gov_0
gov_1
hot_0
hot_1
mfld_0
mfld_1
mus_0
mus_1
nar_0
nar_1
pr_0
pr_1
qas_0
qas_1
natural
needle
0
8
16
24
32
layer
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
far proxy (log)